

Kavyansh Vats

+91 8709065562 | Kavyansh1509@gmail.com | linkedin.com/in/kavyansh-vats | github.com/Kavyansh180

Education

Vellore Institute of Technology | 9.16 CGPA

Bachelor of Technology in Computer Science and Engineering (Core)

Shore, India

Sept 2023 – July 2027

Technical Skills

Programming Languages: Java, Python, SQL

Developer Tools & Technologies: MySQL, REST APIs, GitHub, Data Structures and Algorithms, CI/CD fundamentals, Object-Oriented Programming

AWS Cloud: EC2, S3, Lambda, CloudFront, Load Balancing, Queue

Projects

CustemAI: Hyper-Personalized Marketing Engine | *ReactJS, Tailwind* | [Github](#) | [Demo](#) June 2026 – July 2026

- Addressed the inefficiency of generic marketing campaigns by targeting the need for dynamically generated, hyper-personalized multi-channel content (Email, Push, Display).
- Engineered a real-time LLM-based agent (ReactJS, Groq API/LLaMA 3.3 70B) utilizing prompt engineering and built predictive engagement scoring with micro-persona targeting.
- Generated structured marketing JSON payloads with sub-second response latency for seamless downstream execution in enterprise systems like Epsilon's COREai.

Autonomous Traffic Flow Optimizer | *Python, PyTorch, RL Agent* | [Github](#) | [Demo](#) May 2026 – June 2026

- Identified the limitations of static traffic management by targeting the need for adaptive, data-driven intersection control to minimize vehicle wait times and reduce congestion.
- Trained a PPO reinforcement learning agent on a custom Gymnasium environment (200K timesteps) and deployed it as a serverless REST API utilizing a scalable cloud-native AWS architecture (FastAPI, Docker, Lambda, ECR).
- Cut average queue length by 14.3% compared to fixed-timer baselines, enabling real-time stateful closed-loop control continuous optimization via a Streamlit dashboard.

Food Delivery System (LLD) | *Java, OOP, Design Patterns* | [Github](#) June 2026 – July 2026

- Identified the challenges of inflexible, tightly coupled food delivery platforms by targeting the need for a scalable, reliable, and modular low-level system design.
- Engineered a robust backend architecture in Java applying SOLID principles, implementing the Strategy pattern for multi-mode search and nearest-agent assignment, alongside the Observer pattern for real-time notifications.
- Delivered a highly extensible system foundation capable of managing large-scale operations, ensuring the business logic adapts seamlessly and dynamically to evolving requirements.

Achievements & Certifications

AWS Certified Solutions Architect: Validated expertise in designing distributed systems, showcasing proficiency in scalable cloud architecture, secure infrastructure deployment, and cost-optimized solutions within the AWS ecosystem. [Link](#)

AWS Certified AI Practitioner: Demonstrated basic understanding of artificial intelligence, machine learning, and generative AI concepts, including practical model implementation and ethical considerations using AWS services. [Link](#)

Bronze Medalist (Google Olympics): Placed in The Ultimate Tech and Sports Challenge at VIT Bhopal, demonstrating competitive problem-solving and technical agility. [Link](#)

Extracurricular Activities

Event Anchor, Ekfraseis Dramatics Society

Rangmanch 2025 - Flagship Cultural Event

2025

Bhopal, India

- Hosted the flagship cultural event, taking ownership of the program flow and engaging an audience of 500+ attendees throughout the show.
- Coordinated seamless transitions between performances, ensuring consistent pacing and audience engagement in a collaborative environment.